

HUDSON BAY, SK, Canada

WMO: 718680

Lat: 52.814N

Lon: 102.309W

Elev: 358

StdP: 97.10

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.9	(c) -31.1	(d) -38.2	(e) 0.1	(f) -33.7	(g) -35.5	(h) 0.1	(i) -30.9	(j) 9.7	(k) -12.3	(l) 8.5	(m) -12.1	(n) 2.1	(o) 210	(p) 0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	28.6	19.0	26.9	18.2	25.3	17.3	20.8	26.6	19.5	24.8	18.4	23.6	3.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.2	23.6	17.5	13.1	22.0	16.4	12.2	21.0	62.0	26.8	57.4	24.9	53.8	23.6	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.8	(c) 6.9	(e) -38.4	(f) 32.2	(g) 2.4	(h) 1.8	(i) -40.1	(j) 33.5	(k) -41.6	(l) 34.5	(m) -42.9	(n) 35.6	(o) -44.7	(p) 36.9		
(a) 9.0	(b) 7.8	(c) 6.9	(e) -38.5	(f) 22.9	(g) 2.4	(h) 1.8	(i) -40.2	(j) 24.2	(k) -41.6	(l) 25.3	(m) -43.0	(n) 26.3	(o) -44.7	(p) 27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-16.9	-14.6	-7.4	2.1	9.3	14.7	17.8	16.4	11.1	3.4	-6.4	-14.4		
	DBStd	13.71	8.60	7.82	7.85	6.14	4.88	3.80	3.09	3.43	4.39	4.79	6.84	8.21		
	HDD10.0	3877	833	688	539	242	74	6	0	1	37	209	493	755		
	HDD18.3	6259	1091	922	797	486	282	118	48	79	218	462	743	1013		
	CDD10.0	724	0	0	0	6	52	148	241	199	72	7	0	0		
	CDD18.3	64	0	0	0	0	2	11	30	19	3	0	0	0		
	CDH23.3	859	0	0	0	3	43	146	338	265	63	2	0	0		
	CDH26.7	179	0	0	0	1	7	32	68	60	12	0	0	0		
(14) Wind	WSAvg	3.1	3.0	3.1	3.3	3.5	3.6	3.0	2.8	2.7	3.0	3.1	3.0	2.9		
(15) Precipitation	PrecAvg	479	20	16	28	27	43	72	81	59	57	30	23	24		
	PrecMax	604	45	33	57	73	107	178	150	121	104	84	60	51		
	PrecMin	291	2	4	7	4	10	14	17	6	7	8	5	14		
	PrecStd	83	11	8	14	20	29	40	36	33	30	21	13	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.9	10.9	22.4	27.5	30.0	30.8	31.2	28.2	21.5	11.2	4.4		
		MCWB	1.7	1.7	5.5	11.9	15.2	19.5	20.7	20.7	17.5	13.1	6.4	1.6		
	2%	DB	1.0	1.9	8.2	18.6	24.2	27.0	28.5	28.2	25.1	16.9	7.1	1.2		
		MCWB	-0.6	0.0	3.7	9.4	13.2	17.8	19.5	19.2	16.1	10.3	3.8	-0.4		
	5%	DB	-1.7	-0.6	5.6	15.5	21.7	24.7	26.9	26.3	22.2	13.8	4.4	-1.5		
		MCWB	-2.8	-2.1	2.2	7.8	11.9	16.4	18.8	18.3	14.9	8.7	2.0	-2.7		
	10%	DB	-4.2	-3.1	3.1	12.3	19.5	22.7	25.2	24.4	19.5	11.1	2.2	-3.5		
		MCWB	-5.1	-4.3	0.7	6.2	11.0	15.2	18.0	17.3	13.4	7.2	0.6	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.1	5.7	12.2	16.3	21.4	22.6	22.1	18.9	13.7	6.7	1.9		
		MCDB	4.1	4.1	10.0	21.3	23.2	27.9	27.9	28.4	25.5	20.1	10.5	4.0		
	2%	WB	-0.3	-0.2	3.9	10.0	14.5	19.0	21.1	20.5	17.3	11.3	4.1	-0.3		
		MCDB	0.9	1.7	7.6	17.2	22.0	24.6	26.6	26.5	23.4	15.8	6.6	1.3		
	5%	WB	-2.8	-2.1	2.3	8.2	13.1	17.7	19.8	19.2	15.6	9.3	2.2	-2.6		
		MCDB	-1.7	-0.8	5.3	14.7	20.0	23.0	24.9	24.3	20.8	13.2	4.0	-1.6		
	10%	WB	-5.2	-4.3	0.8	6.5	11.7	16.3	18.7	18.1	14.1	7.6	0.6	-4.3		
		MCDB	-4.3	-3.2	3.0	11.7	18.0	21.2	23.8	23.0	18.8	10.8	2.2	-3.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	11.1	11.7	12.4	13.4	12.2	12.2	12.6	12.4	9.8	8.2	8.4		
		MCDBR	9.8	11.2	12.7	17.5	17.3	15.4	14.8	15.9	17.5	15.3	10.7	9.4		
	5% WB	MCWBR	8.5	9.3	8.8	9.9	8.8	7.5	7.3	7.9	9.4	9.2	7.6	8.0		
		MCWBR	9.7	10.7	12.0	16.1	14.9	13.7	13.1	14.0	15.9	14.3	9.9	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.253	0.280	0.337	0.349	0.374	0.378	0.367	0.329	0.290	0.256	0.232		
	taud		2.304	2.314	2.353	2.382	2.416	2.370	2.389	2.420	2.516	2.580	2.488	2.347		
	Ebn at Noon		769	877	919	899	901	877	867	858	857	832	765	725		
	Edh at Noon		58	79	95	106	108	115	111	102	82	62	49	46		
(44) All-Sky Solar Radiation	RadAvg		1.07	2.19	3.73	4.93	5.62	5.67	5.90	4.85	3.42	1.96	0.95	0.75		
	RadStd		0.08	0.15	0.25	0.39	0.50	0.43	0.33	0.42	0.34	0.28	0.13	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	-0.49	-0.80	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page